

GRAPH NEURAL NETWORKS



Can we trust this person?



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

Can we trust this person?



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

On his own, nothing looks suspicious.

Can we trust this person?

PITCH STAFF



*ordered to
mop the pitch*



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

On his own, nothing looks suspicious.

Can we trust this person?

PITCH STAFF



*ordered to
mop the pitch*

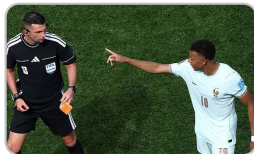


KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

*told how to
referee*

THE REFEREE



On his own, nothing looks suspicious.

Can we trust this person?

PITCH STAFF



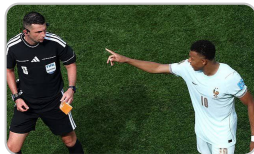
*ordered to
mop the pitch*



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

THE REFEREE



*told how to
referee*

*stared down
through the screen*



YOU, THE VIEWER

On his own, nothing looks suspicious.

Can we trust this person?

PITCH STAFF



*ordered to
mop the pitch*



KYLIAN MBAPPÉ

FORWARD · WORLD CUP WINNER

THE REFEREE



*told how to
referee*

*stared down
through the screen*



YOU, THE VIEWER

Wait... why is everyone taking orders from this guy?

Can we trust this person?



INFERRED FROM HIS CONNECTIONS

DICTATOR?

(OF THE PENALTY BOX)

Can we trust this person?

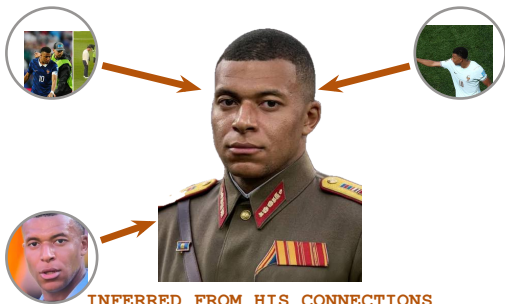


DICTATOR?

(OF THE PENALTY BOX)

What if the person itself
isn't the information?

Can we trust this person?



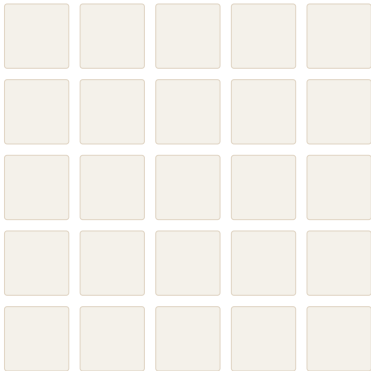
INFERRED FROM HIS CONNECTIONS

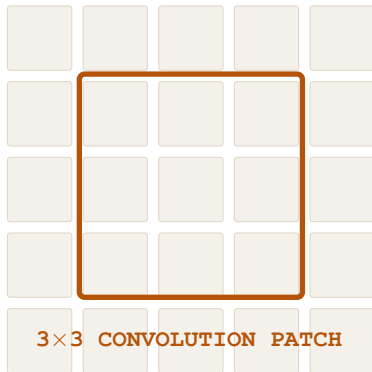
DICTATOR?

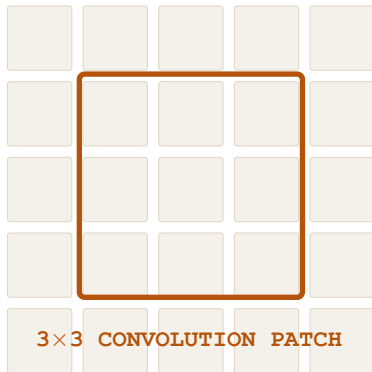
(OF THE PENALTY BOX)

What if the person itself
isn't the information?

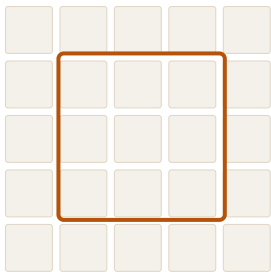
What if the
connections are?







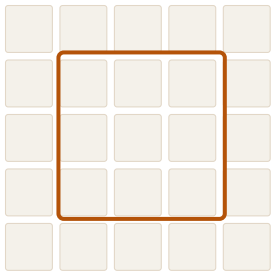
LOCAL CONNECTIVITY · FIXED SPATIAL GRID



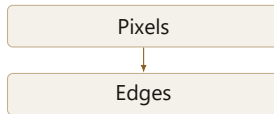
LOCAL PATCHES

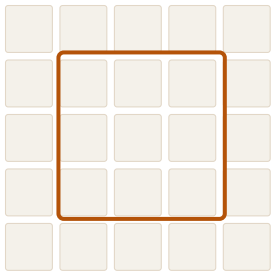


Pixels

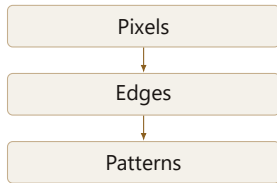


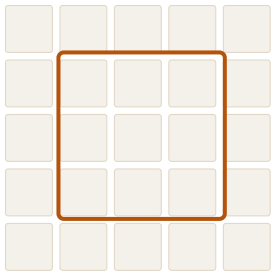
LOCAL PATCHES



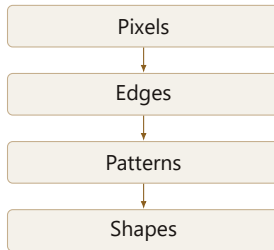


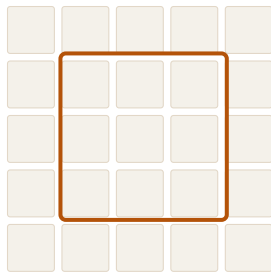
LOCAL PATCHES



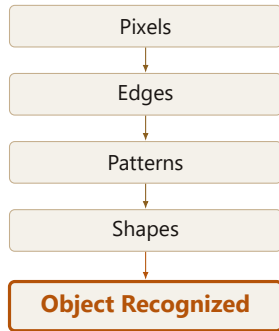


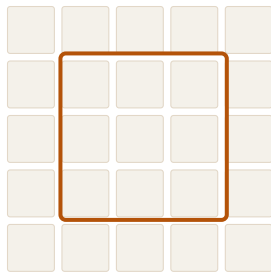
LOCAL PATCHES



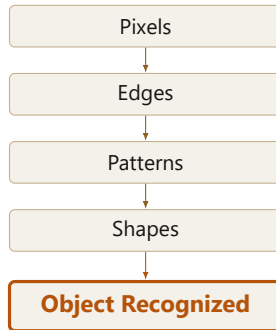


REGULAR PIXEL GRID





REGULAR PIXEL GRID



RIGID GEOMETRIC HIERARCHY

Convolutional Neural Network

Convolutional Neural Network

CNN

Convolutional Neural Network

CNN

BUILT FOR REGULAR EUCLIDEAN GRIDS

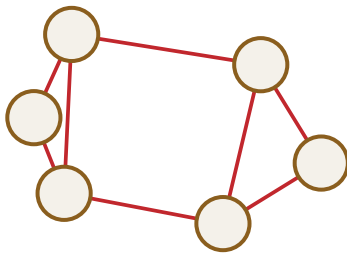




What happens when the
data has no grid?



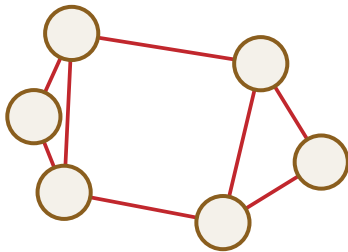
REGULAR GRID



IRREGULAR GRAPH



REGULAR GRID

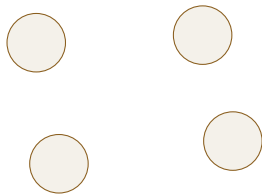


IRREGULAR GRAPH

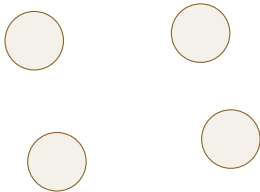
There is no pixel 37.

NO FIXED ORDER · ARBITRARY NEIGHBORS

**SOLUTION 1:
DISCARD THE EDGES**



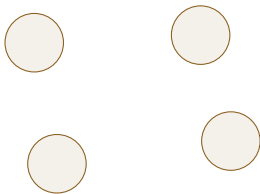
**SOLUTION 1:
DISCARD THE EDGES**



LOSES GRAPH STRUCTURE

What if we keep them?

**SOLUTION 1:
DISCARD THE EDGES**



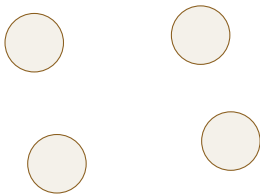
LOSES GRAPH STRUCTURE

**SOLUTION 2:
FLATTEN THE ADJACENCY MATRIX**

$A \rightarrow [0, 1, 0, 1, 1, 0, 1, 0, \dots]$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

SOLUTION 1:
DISCARD THE EDGES



LOSES GRAPH STRUCTURE

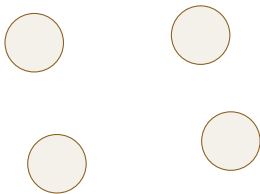
SOLUTION 2:
FLATTEN THE ADJACENCY MATRIX

$A \rightarrow [0, 1, 0, 1, 1, 0, 1, 0, \dots]$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

SENSITIVE TO NODE ORDERING

SOLUTION 1:
DISCARD THE EDGES



LOSES GRAPH STRUCTURE

SOLUTION 2:
FLATTEN THE ADJACENCY MATRIX

$A \rightarrow [0, 1, 0, 1, 1, 0, 1, 0, \dots]$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

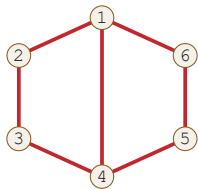
SENSITIVE TO NODE ORDERING

Why do naive approaches fail?

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

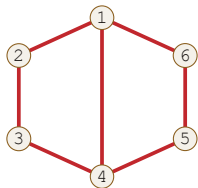
NUMBERING A



TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

NUMBERING A



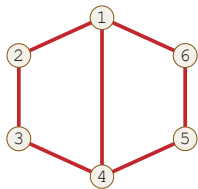
Toxicity: 0.83

ILLUSTRATIVE PREDICTION

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

NUMBERING A



Toxicity: 0.83

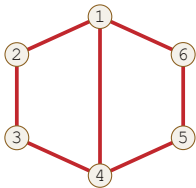
ILLUSTRATIVE PREDICTION

What if we simply
renumber the atoms?

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

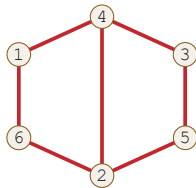
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

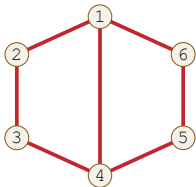
NUMBERING B



TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

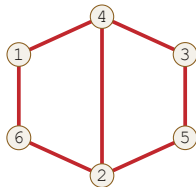
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

NUMBERING B



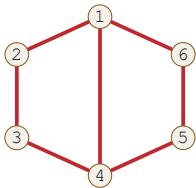
Toxicity: 0.11

ILLUSTRATIVE PREDICTION

TASK: PREDICT WHETHER THIS MOLECULE IS TOXIC

Same molecule. Same atoms. Same bonds.

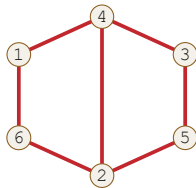
NUMBERING A



Toxicity: 0.83

ILLUSTRATIVE PREDICTION

NUMBERING B



Toxicity: 0.11

ILLUSTRATIVE PREDICTION

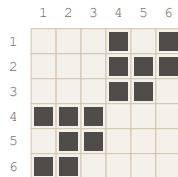
PERMUTATION SENSITIVITY · ARBITRARY NUMBERING BIAS

NUMBERING A · ADJACENCY MATRIX



Toxicity: 0.83

NUMBERING B · ADJACENCY MATRIX



Toxicity: 0.11

SAME BONDS
DIFFERENT FILLING

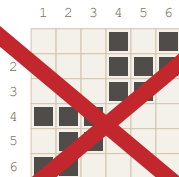
NUMBERING A · ADJACENCY MATRIX



Toxicity: 0.83

SAME BONDS
DIFFERENT FILLING

NUMBERING B · ADJACENCY MATRIX



Toxicity: 0.11

NUMBERING A · ADJACENCY MATRIX



Toxicity: 0.83

NUMBERING B · ADJACENCY MATRIX



Toxicity: 0.11

SAME BONDS
DIFFERENT FILLING

The network learned our filing system.

NUMBERING A · ADJACENCY MATRIX



Toxicity: 0.83

NUMBERING B · ADJACENCY MATRIX



Toxicity: 0.11

SAME BONDS
DIFFERENT FILLING

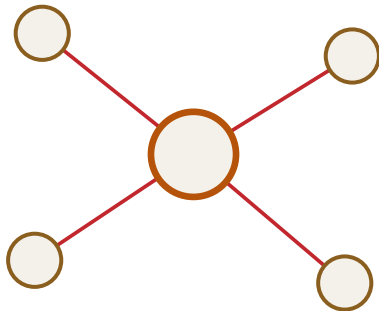
The network learned our filing system.
It never learned the chemistry.

What is this node?



What **is** this node?

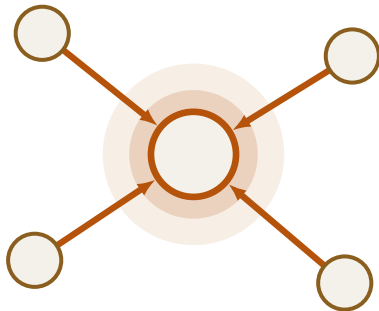
What **is** connected to it?



What **is** this node?

What is connected to it?

Who influences this node?

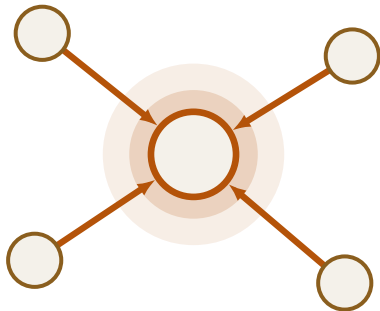


What **is** this node?

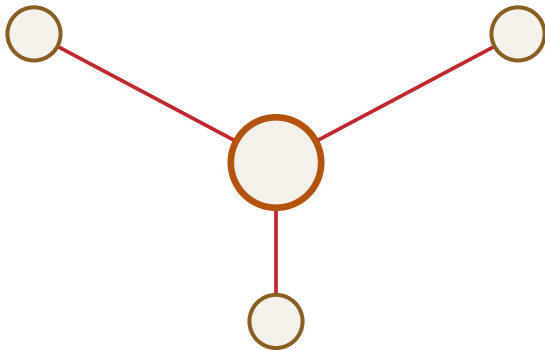
What is connected to it?

Who influences this node?

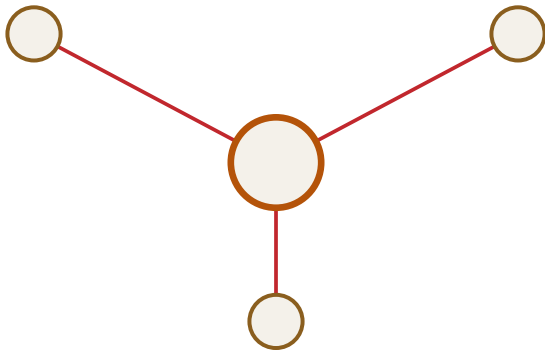
RELATIONAL REASONING



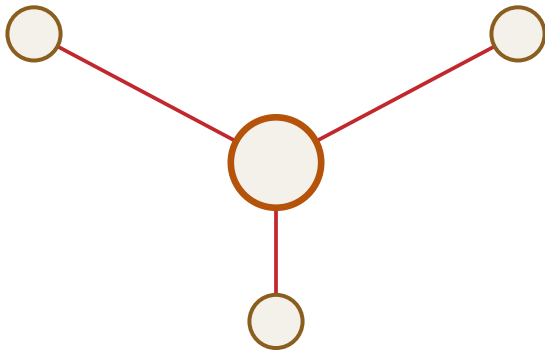
We need to use the neighbourhood.



That is the idea behind a **Graph Neural Network**.

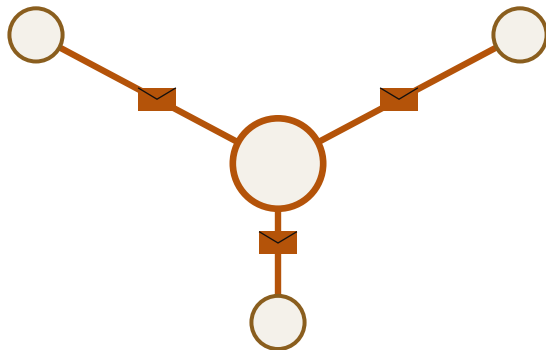


It passes messages between connected nodes.



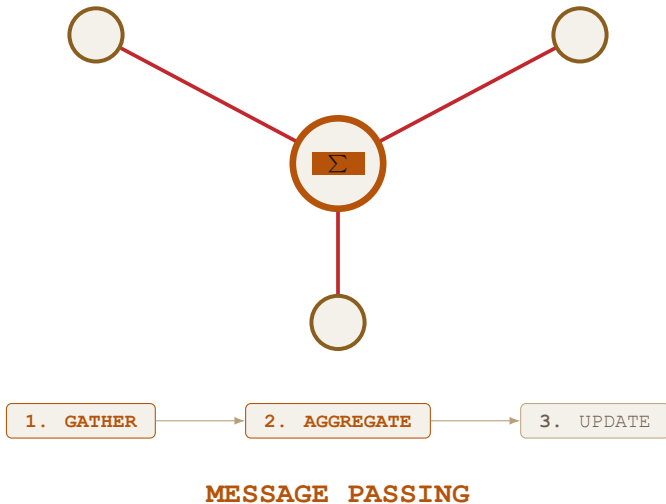
MESSAGE PASSING

It passes messages between connected nodes.

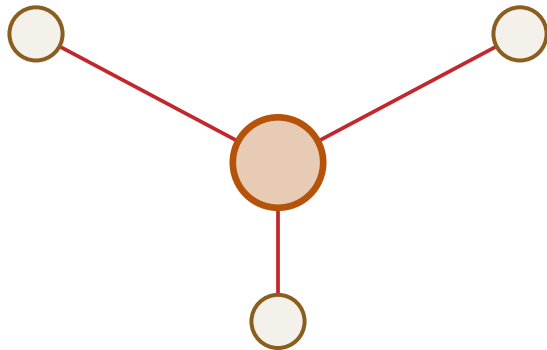


MESSAGE PASSING

It passes messages between connected nodes.

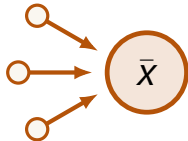


It passes messages between connected nodes.



MESSAGE PASSING

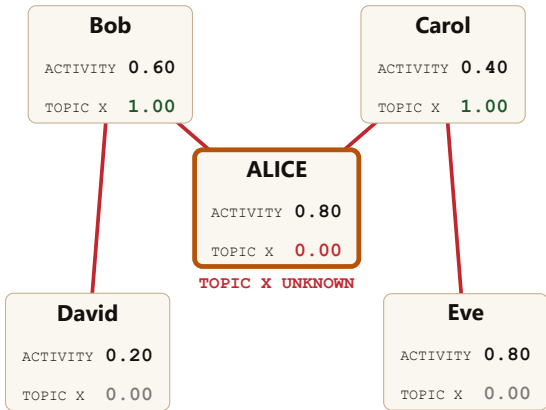
What went wrong, and how a **GNN** fixes it



Answer depends on numbering

Order doesn't matter

Is Alice likely to be interested in Topic X?



SLOT 1 ACTIVITY

how much they post

0 = never 1 = constantly

SLOT 2 TOPIC X

how much they engage with Topic X

0 = never 1 = all the time

THE FIVE CARDS

Bob [0.60 , 1.00]

Carol [0.40 , 1.00]

David [0.20 , 0.00]

Eve [0.80 , 0.00]

Alice [0.80 , 0.00]

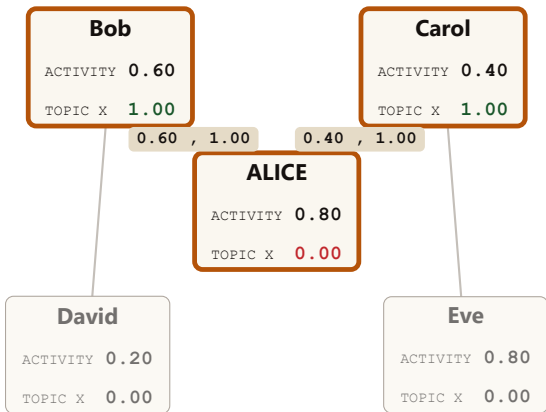
1. GATHER

2. AGGREGATE

3. UPDATE

= ONE LAYER

Step 1 of 3 — GATHER



STEP 1 GATHER

$N(\text{Alice}) = \{ \text{Bob} , \text{Carol} \}$

WHAT IS A MESSAGE?

Bob sends [0.60 , 1.00]

Carol sends [0.40 , 1.00]

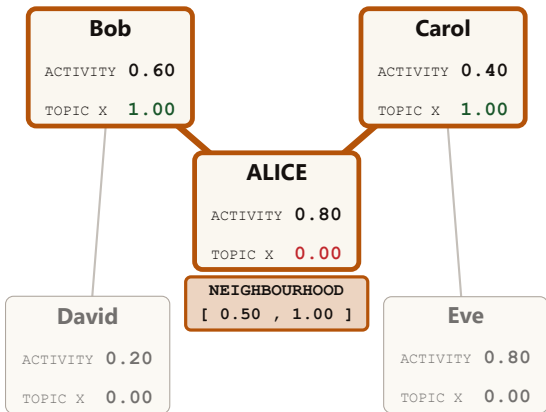
Gathering = Copying messages

1. GATHER

2. AGGREGATE

3. UPDATE

Step 2 of 3 — AGGREGATE



STEP 2 AGGREGATE

Bob [0.60 , 1.00]

Carol [0.40 , 1.00]

Rule: average them, slot by slot.

ACTIVITY $(0.60 + 0.40) / 2 = 0.50$

TOPIC X $(1.00 + 1.00) / 2 = 1.00$

$m(\text{Alice}) = [0.50 , 1.00]$

ORDER INVARIANCE

$\text{mean}(\text{Bob}, \text{Carol}) = \text{mean}(\text{Carol}, \text{Bob})$

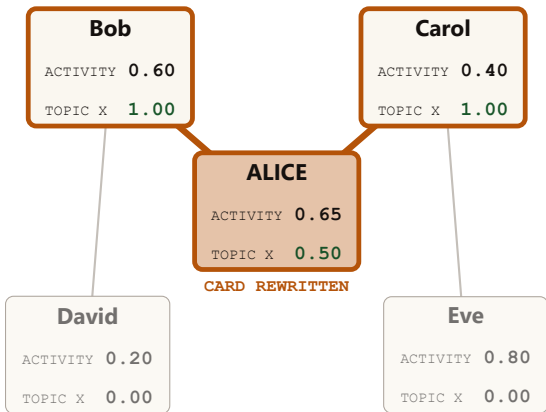
Order Invariant Aggregation

1. GATHER

2. AGGREGATE

3. UPDATE

Step 3 of 3 — UPDATE



STEP 3 UPDATE

mine [0.80 , 0.00]

theirs [0.50 , 1.00]

new = 0.5 x mine + 0.5 x theirs

ACTIVITY $0.5(0.80) + 0.5(0.50)$
 $= 0.40 + 0.25 = 0.65$

TOPIC X $0.5(0.00) + 0.5(1.00)$
 $= 0.00 + 0.50 = 0.50$

$h'(Alice) = [0.65 , 0.50]$

1. GATHER

2. AGGREGATE

3. UPDATE

One blank slot, filled in by the neighbourhood.



READOUT LAYER

$$p = \sigma(6 \times 0.50 - 0.9) = \sigma(2.1) = 0.89$$

LIKELY INTERESTED 89%

PREDICTION DERIVED FROM CONNECTIONS

Alice was not special. Every node ran the same three steps.



ALL NODES CONVERGE TOWARD 0.50 IN SLOT 2

POTENTIAL BOTTLENECK: OVER-SMOOTHING

What the **neighbours** contribute

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

how much of them
 0.5

the shuffle-proof combiner
 average = [0.50 , 1.00]

each neighbour's current card
 Bob = [0.60 , 1.00]

my neighbours
 { Bob , Carol }

Gather · Aggregate

What **Alice** keeps, and the new card

squash it back into range how much of me

$0.5 \times [0.80, 0.00]$

$$h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$$

Alice's new card
[0.65 , 0.50]

The diagram illustrates the calculation of the next hidden state $h_v^{(k+1)}$. It shows the equation $h_v^{(k+1)} = \sigma \left(W_{\text{self}} h_v^{(k)} + W_{\text{nbr}} \text{AGG} \left(\{ h_u^{(k)} : u \in \mathcal{N}(v) \} \right) \right)$. Annotations include: 'squash it back into range' pointing to the σ function; 'how much of me' pointing to the W_{self} weight; and 'Alice's new card [0.65 , 0.50]' pointing to the $h_v^{(k+1)}$ result. A calculation $0.5 \times [0.80, 0.00]$ is shown above the equation, with an arrow pointing to the W_{nbr} weight.

Gather · Aggregate · Update

From Message Passing to Predictions



k-HOP NEIGHBORHOOD EMBEDDINGS

Scenario: Classifying the Dhaka Suspect

SUSPECT EMBEDDING h_{target}

Target #4091

Scenario: Classifying the Dhaka Suspect

SUSPECT EMBEDDING h_{target}

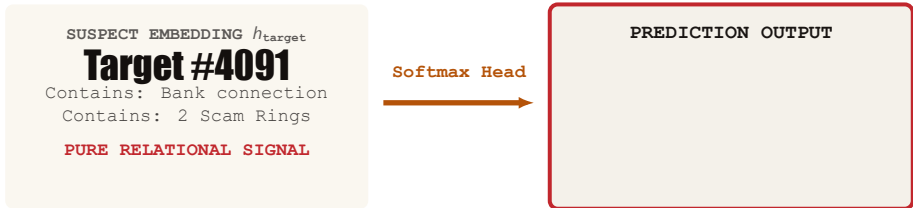
Target #4091

Contains: Bank connection

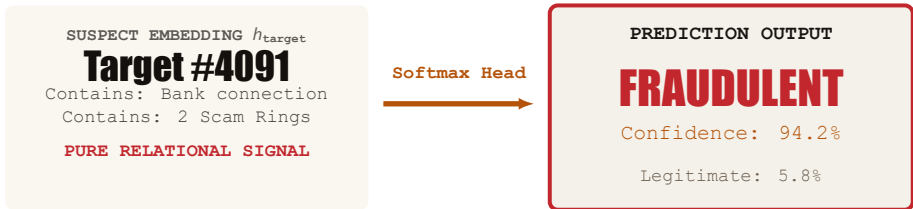
Contains: 2 Scam Rings

PURE RELATIONAL SIGNAL

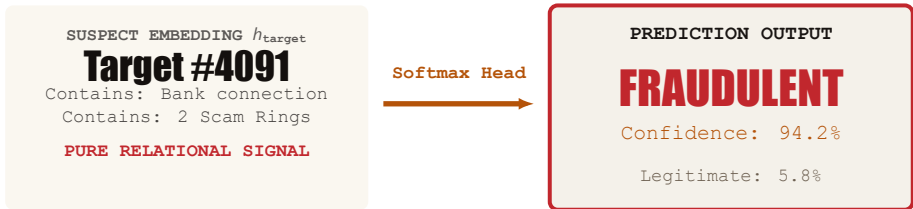
Scenario: Classifying the Dhaka Suspect



Scenario: Classifying the Dhaka Suspect



Scenario: Classifying the Dhaka Suspect



GUILT BY ASSOCIATION · LEARNED MATHEMATICALLY

NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$

THE THREE GRAPH PREDICTION TASKS

NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$

EDGE LEVEL

Link Prediction

Will these two nodes
become connected?

$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$

THE THREE GRAPH PREDICTION TASKS

NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$

EDGE LEVEL

Link Prediction

Will these two nodes
become connected?

$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$

GRAPH LEVEL

Property Test

Is this entire molecule
toxic or curative?

$$h_G = \text{Pool}(\{h_v\})$$

THE THREE GRAPH PREDICTION TASKS

NODE LEVEL

Classification

Is this account
a scammer?

$$\hat{y}_v = \text{Softmax}(Wh_v)$$

EDGE LEVEL

Link Prediction

Will these two nodes
become connected?

$$\hat{y}_{uv} = \sigma(h_u^T h_v)$$

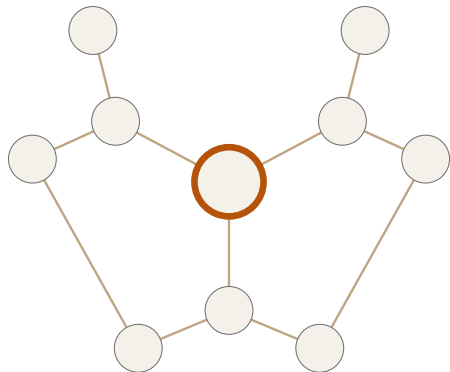
GRAPH LEVEL

Property Test

Is this entire molecule
toxic or curative?

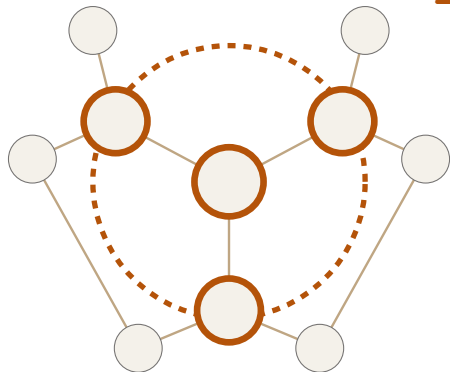
$$h_G = \text{Pool}(\{h_v\})$$

ONE MECHANISM · THREE PREDICTION SCALES

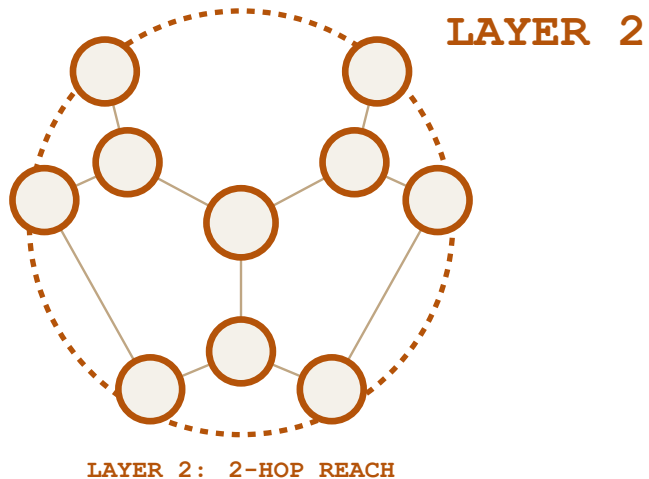


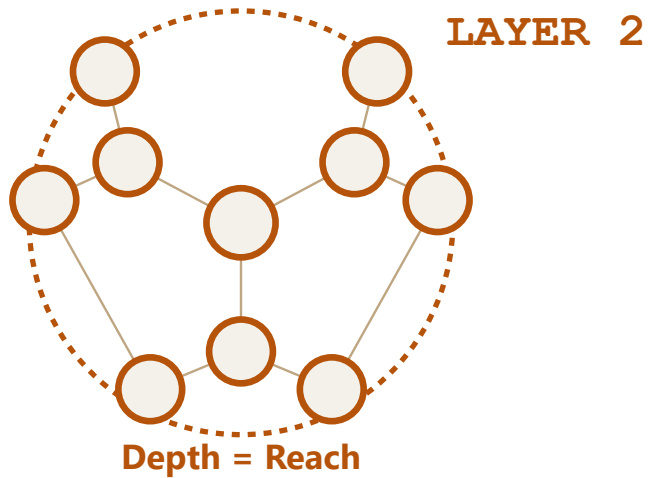
INPUT GRAPH: 1-HOP NEIGHBORHOOD

LAYER 1

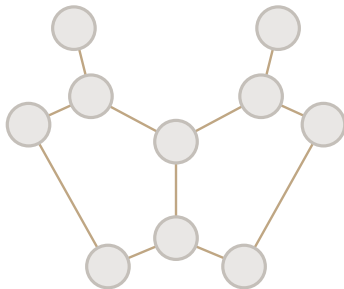


LAYER 1: 1-HOP REACH



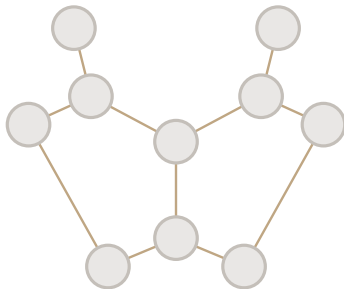


LAYER 6



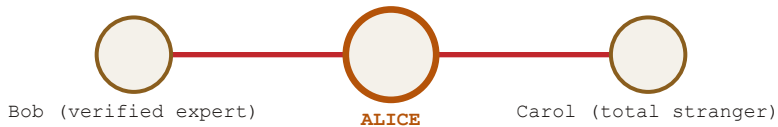
OVER-SMOOTHING

LAYER 6

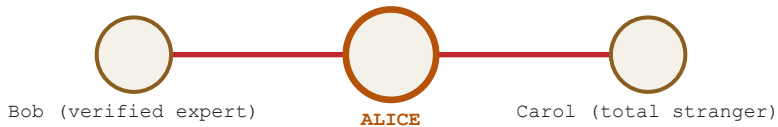


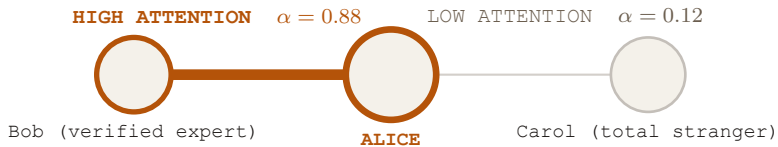
OVER-SMOOTHING
CONVERGENCE TO MEAN · LIMIT: 2-3 LAYERS

We averaged Bob and Carol equally. Should we have?

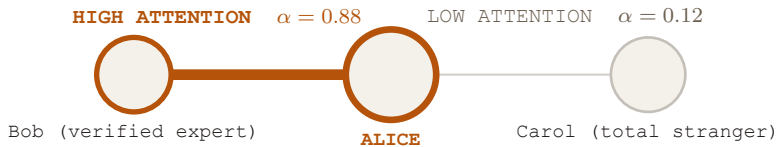


We averaged Bob and Carol equally. Should we have?





LEARNED ATTENTION WEIGHTS (α_{ij})



LEARNED ATTENTION WEIGHTS (α_{ij})

DYNAMIC PER-EDGE ATTENTION COEFFICIENTS

GCN

Average everyone

GCN

Average everyone

GraphSAGE

Sample a few

GCN

Average everyone

GraphSAGE

Sample a few

GAT

Learn who to trust

GCN

Average everyone

GraphSAGE

Sample a few

GAT

Learn who to trust

GATHER · AGGREGATE · ATTEND

GCN

Average everyone

GraphSAGE

Sample a few

GAT

Learn who to trust

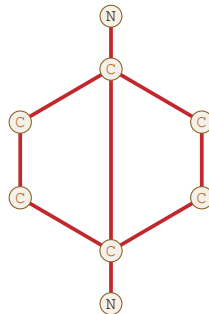
GATHER · AGGREGATE · ATTEND

HOW TO GATHER · HOW TO AGGREGATE · WHO TO LISTEN TO

MIT · 2020

Halicin

HALICIN GRAPH REPRESENTATION

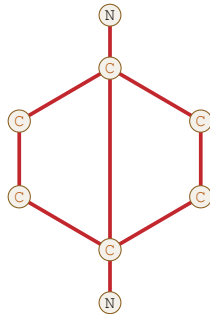


MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.

HALICIN GRAPH REPRESENTATION

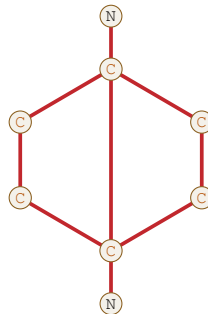


MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.
- Learned chemical bond topologies, not atom numbering.

HALICIN GRAPH REPRESENTATION

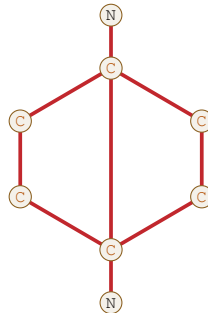


MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.
- Learned chemical bond topologies, not atom numbering.
- Flagged a molecule humans had completely overlooked.

HALICIN GRAPH REPRESENTATION



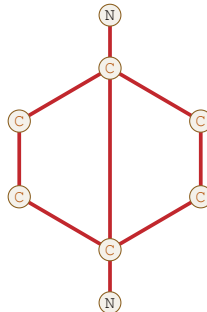
MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.
- Learned chemical bond topologies, not atom numbering.
- Flagged a molecule humans had completely overlooked.

Killed Pan-Resistant Bacteria

HALICIN GRAPH REPRESENTATION



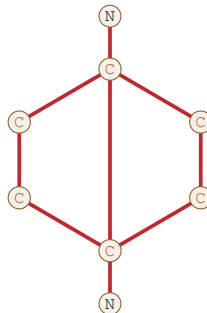
MIT · 2020

Halicin

- Screened **100 million** molecules as graphs.
- Learned chemical bond topologies, not atom numbering.
- Flagged a molecule humans had completely overlooked.

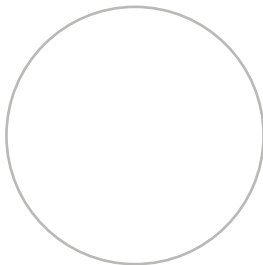
Killed Pan-Resistant Bacteria

HALICIN GRAPH REPRESENTATION

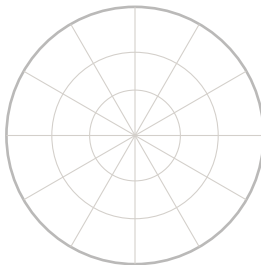


MOLECULAR GRAPH · ANTIBIOTIC DISCOVERY

CASE FILE 02: THE ENTIRE ATMOSPHERE

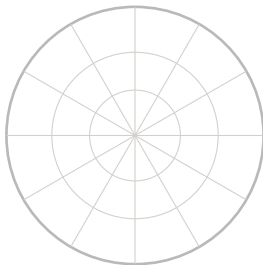


CASE FILE 02: THE ENTIRE ATMOSPHERE



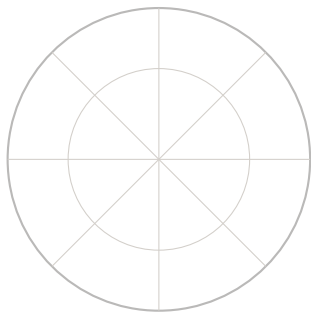
ATMOSPHERIC SPHERICAL MESH GRAPH

CASE FILE 02: THE ENTIRE ATMOSPHERE



ATMOSPHERIC SPHERICAL MESH GRAPH

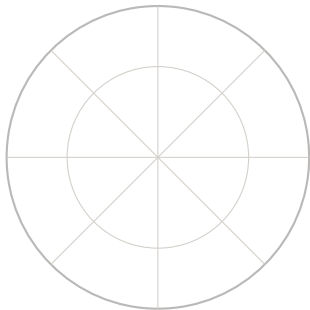
SPHERICAL MESH GRAPH · PLANETARY AIRFLOW



PLANETARY SCALE

DEEPMIND · 2023

GraphCast

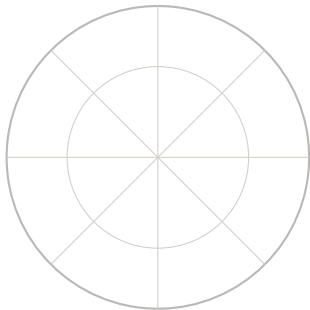


PLANETARY SCALE

DEEPMIND · 2023

GraphCast

- 10-day global forecast in **under 1 minute** on one machine.

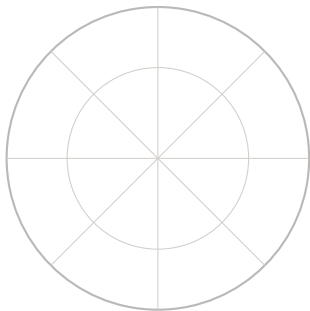


PLANETARY SCALE

DEEPMIND · 2023

GraphCast

- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on >90% of variables.

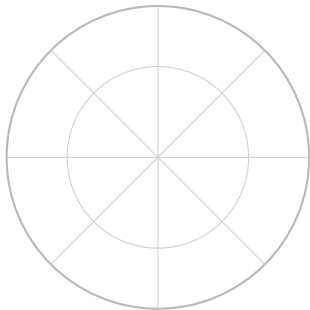


PLANETARY SCALE

DEEPMIND · 2023

GraphCast

- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on >90% of variables.
- Predicted Hurricane Lee days earlier.



PLANETARY SCALE

DEEPMIND · 2023

GraphCast

- 10-day global forecast in **under 1 minute** on one machine.
- Beat the European gold-standard supercomputers on >90% of variables.
- Predicted Hurricane Lee days earlier.

Same Principle. Planetary Scale.

CASE FILE 03: CERN LARGE HADRON COLLIDER



Which hits belong to the same particle?

CERN LHC · CMS / ATLAS

Connect Dots



CERN LHC · CMS / ATLAS

Connect Dots

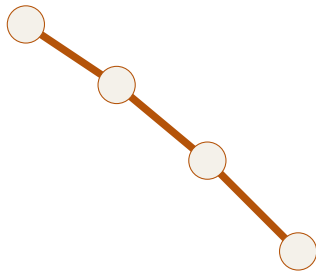
- Builds a graph of candidate hits.



CERN LHC · CMS / ATLAS

Connect Dots

- Builds a graph of candidate hits.
- The GNN prunes false connections in **milliseconds**.

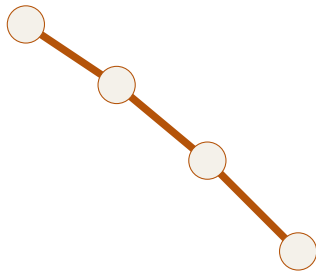


TRUE PARTICLE TRAJECTORY

CERN LHC · CMS / ATLAS

Connect Dots

- Builds a graph of candidate hits.
- The GNN prunes false connections in **milliseconds**.
- Reconstructs particle trajectories in real time.



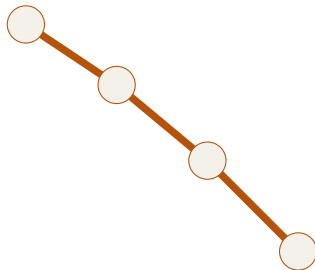
TRUE PARTICLE TRAJECTORY

CERN LHC · CMS / ATLAS

Connect Dots

- Builds a graph of candidate hits.
- The GNN prunes false connections in **milliseconds**.
- Reconstructs particle trajectories in real time.

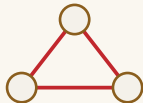
Discovering New Physics



TRUE PARTICLE TRAJECTORY

MOLECULES

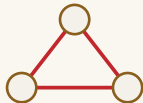
Atoms + Bonds



CHEMISTRY

MOLECULES

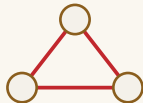
Atoms + Bonds



CHEMISTRY

PLANET

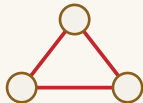
Grid + Airflow



ATMOSPHERE

MOLECULES

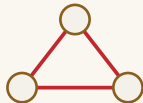
Atoms + Bonds



CHEMISTRY

PLANET

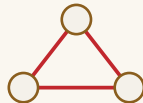
Grid + Airflow



ATMOSPHERE

PARTICLES

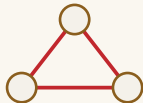
Hits + Tracks



PHYSICS

MOLECULES

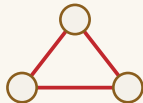
Atoms + Bonds



CHEMISTRY

PLANET

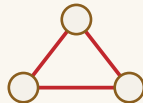
Grid + Airflow



ATMOSPHERE

PARTICLES

Hits + Tracks



PHYSICS

DIVERSE DOMAINS · ONE GRAPH LANGUAGE

GRAPH

GRAPH

Different problems. Same underlying structure.

GRAPH

Different problems. Same underlying structure.

RELATIONSHIPS AS FIRST-CLASS CITIZENS

Thank you.

QUESTIONS & DISCUSSION